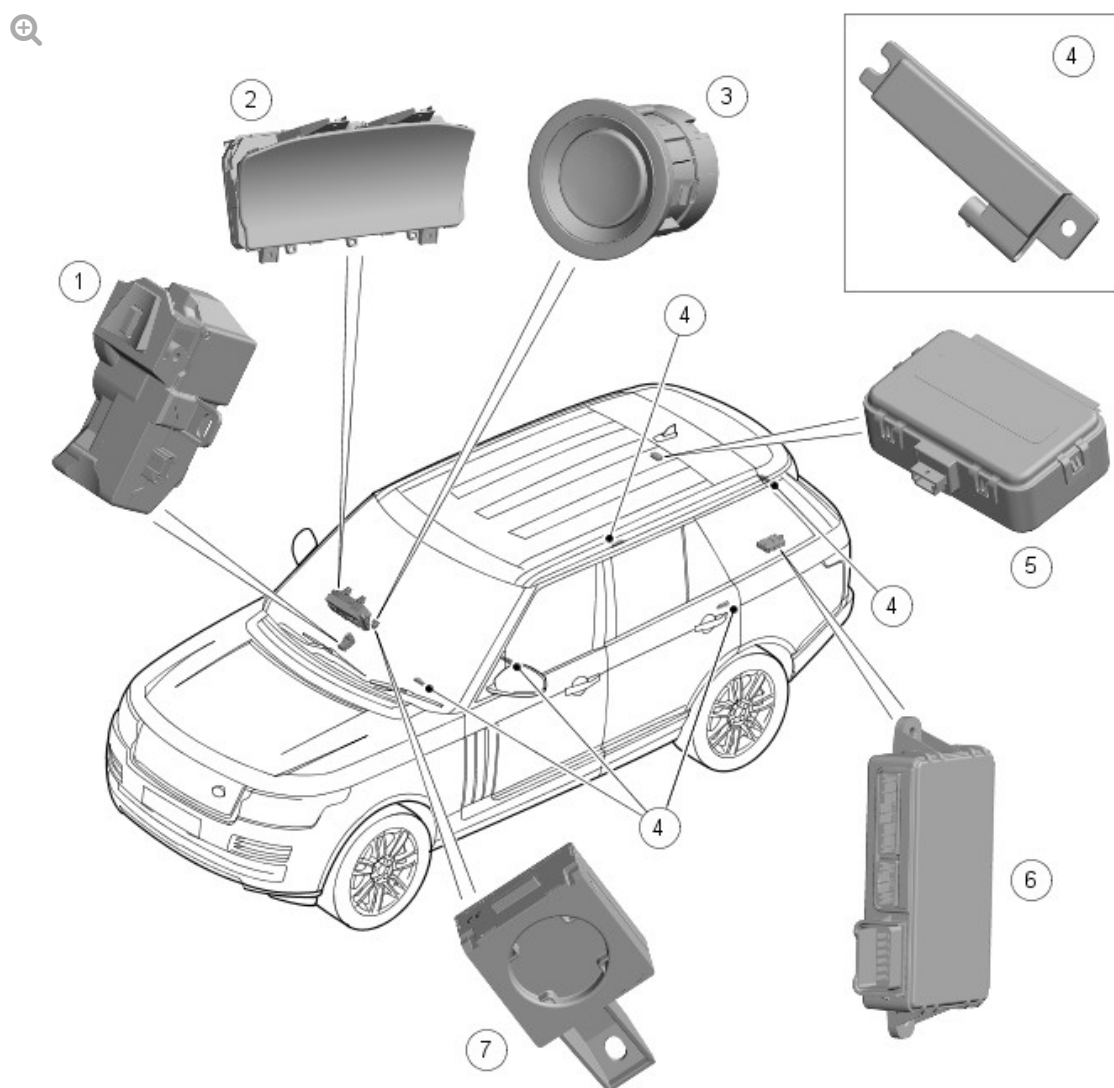


2016.0 RANGE ROVER (LG), 419-01

ANTI-THEFT - PASSIVE

DESCRIPTION AND OPERATION

COMPONENT LOCATION



E149008

ITEM

DESCRIPTION

1	Electric Steering Column Lock (ESCL)
2	Instrument Cluster (IC)
3	Stop/start switch
4	LF antenna (5 off)
5	Radio Frequency (RF) receiver
6	Keyless Vehicle Module (KVM)
7	Immobilizer Antenna Unit (IAU)

OVERVIEW

The passive start system relies on the detection of a uniquely coded smart key via low frequency antennas strategically situated within the vehicle. The antennas ensure the smart key is always within the active transmission zone of the antennas wherever the smart key is placed inside the vehicle. For this reason the orientation and positioning of the antennas is critical to the correct functioning of the system. The smart key also operates the passive entry system.

For additional information, refer to: [Handles, Locks, Latches and Entry Systems](#) (501-14 Handles, Locks, Latches and Entry Systems, Description and Operation).

The system provides a secure interface between the CJB (central junction box) and the ECM (engine control module) to prevent unauthorized starting of the engine. This is achieved by immobilization of the engine crank system and the fuel system, using encoded data exchange between the smart key and multiple control modules.

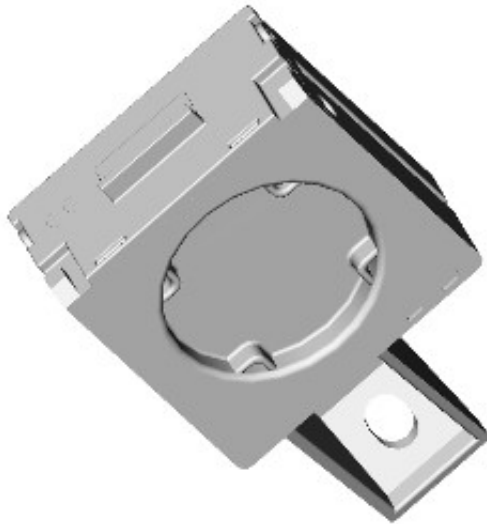
Engine starting is initiated when the encoded data exchange between the smart key and the control modules is verified. The engine management system will then allow engine crank and fueling when an authorization data message is received from the CJB.

The engine can be started by pressing the stop/start switch when 'Park'

position is selected and the brake pedal is pressed.

DESCRIPTION

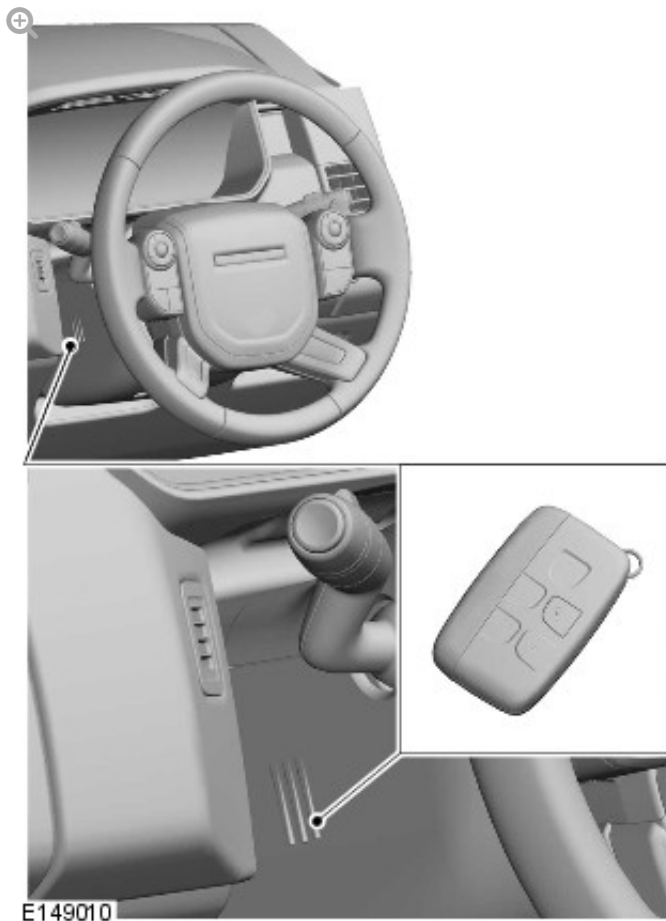
IMMOBILIZER ANTENNA UNIT



E134665

The IAU is used if the KVM is unable to authorize the smart key. If the KVM is unable to identify the smart key, for example if the smart key battery voltage is low or there is local RF interference, the transponder within the smart key can be read in the conventional manner. The driver will be alerted to this by a chime and a message in the IC message center 'SMART KEY NOT FOUND REFER TO HANDBOOK'.

KEYLESS START BACKUP



If the vehicle has been unlocked using the emergency key blade or the smart key is not detected by the vehicle, it will be necessary to use the keyless start backup to disarm the alarm and start the engine. The following process must be followed in this event:

- Press the stop/start switch, if the KVM fails to locate the smart key, the message 'SMART KEY NOT FOUND REFER TO HANDBOOK' shall be displayed.
- Position the smart key against the lower left side of the steering column, with the buttons facing downwards. The ribs denote actual location of the IAU.
- Press the stop/start switch with the brake pedal depressed to start the engine.

NOTE:

If the 'SMART KEY NOT FOUND' message is no longer displayed (only displayed for 10s), then the sequence would have to be repeated.

This process bypasses the data exchange between the KVM and the CJB. This is an inductive process and will operate even if the battery in the smart key is discharged. A transponder within the smart key is detected by the IAU. The IAU communicates this code with the CJB via a LIN (local interconnect network) bus connection. The CJB then initiates the vehicle start process in the normal manner.

LOW FREQUENCY (INTERIOR) ANTENNAS

Five LF (low frequency) antennas for the passive start system are positioned in specific locations within the vehicle.

The KVM transmits an LF signal via the antennas which is received by the smart key. The smart key then responds by transmitting a RF signal which is received by the RF receiver and passed to the KVM for authorization.

KEYLESS VEHICLE MODULE

The KVM controls signal transmissions to and from the smart key and provides authorization to allow the vehicle to be started. The module has a medium speed CAN (controller area network) connection to the CJB for authorizing vehicle starting.

RADIO FREQUENCY RECEIVER

The RF receiver transmission is received from the smart key to enable key identification.

OPERATION

At the request of the CJB the KVM prompts each of the internal low frequency antennas to output a signal. When the smart key is in the vehicle

cabin, it detects the low frequency signals and responds with a RF radio frequency data-identification signal back to the KVM via the RF receiver.

If the data received matches that stored in the KVM, it continues the passive start process by communicating a 'smart key valid' signal to the CJB via the medium speed CAN bus.

Once the CJB receives the authorization and confirms the response with an internal calculation, it passes coded data to the IC on the high speed CAN bus. Upon confirmation from the IC the ignition is enabled.

Before CJB sends a mobilization signal to the ECM, it will exchange encrypted data with the electric steering lock mechanism to authorize unlocking of the steering column. The IC only provides a ground for the steering lock motor.

NOTE:

Electric Steering Column Lock is market dependent.

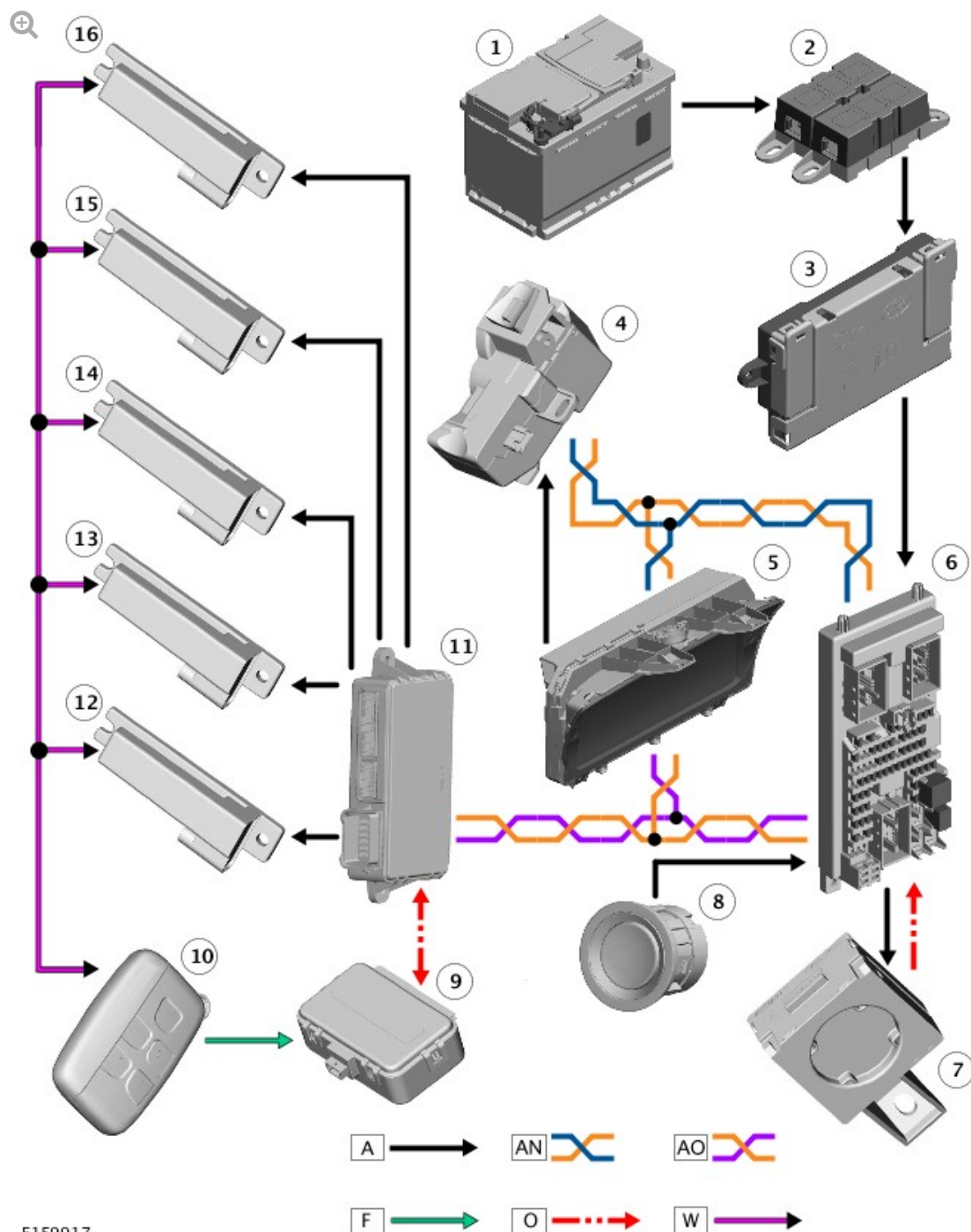
The CJB will enable the fuel pump relay which, on diesel vehicles operates the fuel pump and on gasoline vehicles sends a battery voltage supply to the Fuel Pump Driver Module (FPDM) to operate the fuel pump in conjunction with the ECM.

If the KVM fails to locate the smart key, a message 'SMART KEY NOT FOUND REFER TO HANDBOOK' will appear in the IC message center and the keyless start back-up process will have to be used to mobilize and start the vehicle.

To ensure optimum long term reliability of the smart key the battery must be replaced with a brand new, unused battery. If a used battery is installed the "SMART KEY BATTERY LOW" message may not be cleared. To avoid contamination of the contacts the battery should be removed from its packaging and installed into the smart key while wearing gloves. To confirm

that the replacement battery is working correctly press the unlock button twice while holding the smart key outside the vehicle, then enter the vehicle with the smart key, press the start button and confirm that the "SMART KEY BATTERY LOW" message is not displayed.

CONTROL DIAGRAM



A = HARDWIRED CONNECTION; AN = HIGH SPEED CAN POWERTRAIN SYSTEMS; AO = MEDIUM SPEED CAN BODY SYSTEMS; F = RF TRANSMISSION; O = LIN BUS; W = LF TRANSMISSION.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box (BJB) 2
3	BJB
4	ESCL (where fitted)
5	IC
6	Central Junction Box (CJB)
7	IAU
8	Stop/Start switch
9	RF receiver
10	Smart key
11	KVM
12	LF antenna
13	LF antenna
14	LF antenna
15	LF antenna
16	LF antenna